

# We Built an AI Incident Responder 🚨🤖

Here's What We Got Wrong.

David Pech — Wrike

# Who



David Pech



# The Golden Grail of SRE Agents

## What KubeCon was about

- Every other booth had an "AI SRE"
- Same demo, same Slack screenshot
- Same promise: **no more on-call pages**

## The pitch in one line

Alert fires → agent investigates → agent fixes your precious production

Sounds great. Reality is messier.

# The Golden Grail of SRE Agents

## Reality Check

- 🏷️ Seeing 80% of booths for the last time
- ✍️ Just a few vendors with real production "read-write" experience
- 🛠️ **MCP-everywhere** - just wrapping original API servers oftentimes

## Realistic Pitch

⚠️ **Investigates** = reads dashboards it was told about. New service? It doesn't know.

⚠️ **Fixes** = opens a PR. Someone still merges. Often that's *\*you\**, at 3 AM.

🚫 **Stay asleep** = until the agent confidently does the wrong thing, twice, in prod.

✅ **Realistic win:** cut the "context-gathering" minutes of an incident from 20 → 2.

# Booth promise: the "\$\$\$ profit" demo 💰

🔔 Slack #alerts

PrometheusAlert: payments-api OOMKilled  
namespace=prod, container=app, restarts=7

▼  
you: @agent increase mem for the app

▼  
🤖 agent: opened PR #4172 in gitops/prod

+ memory: 768Mi → 1Gi

+ -Xmx768m → -Xmx896m

▼  
● PR auto-merged

● ArgoCD synced

▼  
💰 \$\$\$ profit

## What this demo silently assumes

📍 The agent already knows **which** GitOps repo, **which** overlay, **which** file

📄 The agent has **write** credentials to a prod repo

🤝 The on-call person **trusts** a one-line Slack request

🧠 +25% memory is actually the right fix (not a leak, not a noisy neighbor)

🔄 No one cycles the agent into a feedback loop with itself

# What we actually shipped

internal "Product" ← the most important part



## Who is it for?

- We are not sure
- Ops + maybe: DevOps, Developers, Managers...

## What should it do?

- We don't know
- Investigate, research, maybe: fix things? Do we really want that?

# Technicals

- Python vibe-coded app on top of Claude Agent SDK
  - With MCPs to PagerDuty, VictoriaMetrics, Netbox, ...
- Tapped into Slack channels (both human and alerting)
- Bundled a lot of our repos
  - Wikis
  - Docs
  - GitOps repos
  - Terraform repos
  - ...
- Workflow: what to do (context)

# Wrong Approach

Vendors are selling the tool: `run agent on your alerts,`

✨ **AI magic will happen™** ✨

Our `ops-responder-agent` :

- could sell you fresh bread over Slack with a different context...
- can be replaced with Gemini Enterprise or any other in a month

Your wrapper around the real agent doesn't matter much and will likely change.

Context and the workflow are.



PART I

# Context, Output & Integration

# What Can You Get Without Context

| <input type="checkbox"/>                                                                            | Status    | Severity                                       | Summary                                                           | Created ↓                | Service                                                                |  |  |  |  |  |  |  |
|-----------------------------------------------------------------------------------------------------|-----------|------------------------------------------------|-------------------------------------------------------------------|--------------------------|------------------------------------------------------------------------|--|--|--|--|--|--|--|
| <input type="checkbox"/>                                                                            | Triggered | Warning                                        | Problem: Cilium agent reports 1 unreachable node(s) in [REDACTED] | [REDACTED] 2026, 9:26 PM | <a href="#">Zabbix Development</a><br><a href="#">Warning severity</a> |  |  |  |  |  |  |  |
| Custom Details                                                                                      |           |                                                |                                                                   |                          |                                                                        |  |  |  |  |  |  |  |
| event_operational_data:                                                                             |           | k8s_cluster: [REDACTED] unreachable_nodes: "1" |                                                                   |                          |                                                                        |  |  |  |  |  |  |  |
| event_severity:                                                                                     |           | Warning                                        |                                                                   |                          |                                                                        |  |  |  |  |  |  |  |
| event_tags:                                                                                         |           |                                                |                                                                   |                          |                                                                        |  |  |  |  |  |  |  |
|                                                                                                     |           | __event_delay:                                 | 5m                                                                |                          |                                                                        |  |  |  |  |  |  |  |
|                                                                                                     |           | __event_recovery_notify:                       | 1                                                                 |                          |                                                                        |  |  |  |  |  |  |  |
|                                                                                                     |           | __metric_source:                               | prometheus                                                        |                          |                                                                        |  |  |  |  |  |  |  |
|                                                                                                     |           | alert:                                         | cilium_unreachable_nodes_max                                      |                          |                                                                        |  |  |  |  |  |  |  |
|                                                                                                     |           | k8s_cluster:                                   | [REDACTED]                                                        |                          |                                                                        |  |  |  |  |  |  |  |
|                                                                                                     |           | scope:                                         | sysops                                                            |                          |                                                                        |  |  |  |  |  |  |  |
|                                                                                                     |           | service_kind:                                  | kubernetes_cni_cilium                                             |                          |                                                                        |  |  |  |  |  |  |  |
|                                                                                                     |           | wiki_page_tag:                                 | kubernetes_cni_cilium                                             |                          |                                                                        |  |  |  |  |  |  |  |
|                                                                                                     |           | zbx_dyntag:                                    | zbx_tag9                                                          |                          |                                                                        |  |  |  |  |  |  |  |
| Zabbix hostname:                                                                                    |           | Service Kubernetes CNI Cilium QA               |                                                                   |                          |                                                                        |  |  |  |  |  |  |  |
| Links: <a href="#">Zabbix Event</a> , <a href="#">Grafana Dashboard</a> , <a href="#">Wiki Docs</a> |           |                                                |                                                                   |                          |                                                                        |  |  |  |  |  |  |  |
| [REDACTED]                                                                                          |           |                                                |                                                                   |                          |                                                                        |  |  |  |  |  |  |  |

# Guess Work

**PagerDuty** APP 17 minutes ago  
**Problem: Cilium agent reports 1 unreachable node(s) in**

Incident type: Base Incident  
Assigned: ZabbixQA common user  
Urgency: Low  
Service: Zabbix Development Warning serverity

[Acknowledge](#) [Create incident channel](#) [SRE agent analysis](#)

More actions ▾

1 reply

**PagerDuty** APP 1 minute ago  
Sent by: PagerDuty Advance SRE Agent

@david.pech, Analysis

Description:

The Cilium CNI agent in the [redacted] Kubernetes cluster is reporting 1 unreachable node, indicating a network connectivity issue within the cluster's CNI layer.

Categorization:

Kubernetes CNI / Network Connectivity — Warning severity, low urgency. Detected via Prometheus metric `cilium_unreachable_nodes_max`.

Key Findings:

- Cluster affected: [redacted]
- Unreachable nodes: 1
- Alert source: Prometheus metric `cilium_unreachable_nodes_max` via Zabbix
- Scope: `sysops` | Service kind: `kubernetes_cni_cilium`
- Alert triggered at: [redacted]
- Recurring pattern: This is a highly recurring alert. There are multiple past similar

Resume: Not bad given how little context it has.

Useful: absolutely not.

@PagerDuty booth discussion on KubeCon NA 2025: "Can we add more context?" "No." (but it might have changed since...)

# You Need to Provide More Context

Our approach:

- mimic the local Claude Code setup
- mono-repo style

Problem:

- CLAUDE.md (AGENTS.md) in each repo
- You should not provide out-of-date, legacy information (**THIS IS BIG**)



**Stale context = confident wrong answers. Pruning beats hoarding.**

# MCP vs. tool use for Wiki

For some reason - MCP significantly less effective.

Medium sized wiki (1000 pages):

- grep and other basic agent tools are very effective

GitOps repos

## Our approach:

- move from some proprietary wiki to Markdown
- we needed to recreate wiki from scratch

Typically 30 - 110 tool uses (Grep|Read|Glob) for a single prompt

~80MB of context, 8MB .tf, 3MB .md

# Session vs. Separate each run

Note: Token caching: 5 mins

## Our approach:

- each @mention is a new session

Trade-off:

- fewer tokens
- more time taken each run
- some details are lost

Surprisingly - not a problem

Example:

```
- Alert A.  
- You: @ops-responder-agent - find the definition of the alert  
- Agent: Definition is ...  
- You: @ops-responder-agent - based on the alert definition  
- Agent: (start from scratch)
```

# Slack Context

Our original idea:

- we absolutely must share the context of all Slack alerting channels, so the agent is aware of the general situation and can leverage it

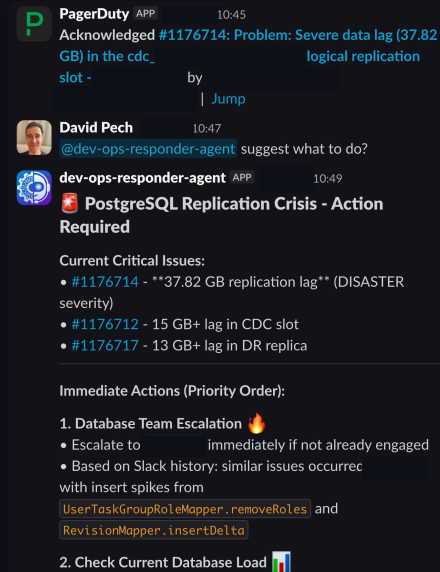
## Workflow

### Step 1: Locate the Prompt Message (MANDATORY)

Before doing ANYTHING else, locate the exact message that triggered this evaluation in `./slack/` history.

### File Structure

```
./slack/  
index.yaml           # Channel list with names and message counts  
{channel_id}/  
  recent.yaml        # Last 24h  
  extended.yaml      # 1-7 days
```



The screenshot shows a Slack thread. At the top, a PagerDuty app message at 10:45 acknowledges a problem: "Severe data lag (37.82 GB) in the cdc\_ logical replication slot -". Below this, David Pech at 10:47 asks the @dev-ops-responder-agent to suggest what to do. The agent's response at 10:49 is titled "PostgreSQL Replication Crisis - Action Required". It lists current critical issues: a disaster-level replication lag of 37.82 GB in channel #1176714, a 15 GB+ lag in CDC slot #1176712, and a 13 GB+ lag in DR replica #1176717. It then lists immediate actions in priority order: 1. Database Team Escalation (with a fire icon), which includes escalating immediately if not engaged, and based on Slack history, inserting spikes from `UserTaskGroupRoleMapper.removeRoles` and `RevisionMapper.insertDelta`; 2. Check Current Database Load (with a server icon).

**PagerDuty** APP 10:45  
Acknowledged #1176714: Problem: Severe data lag (37.82 GB) in the cdc\_ logical replication slot - by | Jump

**David Pech** 10:47  
@dev-ops-responder-agent suggest what to do?

**dev-ops-responder-agent** APP 10:49  
**PostgreSQL Replication Crisis - Action Required**

Current Critical Issues:

- #1176714 - \*\*37.82 GB replication lag\*\* (DISASTER severity)
- #1176712 - 15 GB+ lag in CDC slot
- #1176717 - 13 GB+ lag in DR replica

Immediate Actions (Priority Order):

1. Database Team Escalation 🔥
  - Escalate to immediately if not already engaged
  - Based on Slack history: similar issues occurred with insert spikes from `UserTaskGroupRoleMapper.removeRoles` and `RevisionMapper.insertDelta`
2. Check Current Database Load 🖥️

# Problem: Slack Context

- it uses tools like Grep|Sed - skips the context on purpose



**dev-ops-responder-agent** APP Thread in [redacted]

21:27



Could not locate the triggering message — channel  
C060D8[redacted] is not in Slack history. Answering based on today's  
[redacted] 500-error alerts.



**dev-ops-responder-agent** APP Thread in [redacted] 10:54



**Investigation: app-segment-server Cascading DB Issues**



Could not locate the triggering message in channel  
C146 [redacted] — that channel is not in the available Slack history.  
Investigated all available channels for recent problems and found a  
clear pattern.



# Problem: Context Or Code?

Slack Output, Slack Fetching

Hard to decide, even having both and experimenting between them.

# Slack Output

- no formatting
- have a script replace .md / something to Block Kit
- explain that in a context

SLACK\_OUTPUT.md 11.68 KiB

Code Preview

Slack Block Kit Output Format

When `<output_mode>slack</output_mode>` is present in the prompt, output your response as a JSON array of Block Kit blocks wrapped in `<slack_output>` tags.

Output Format

CRITICAL: Wrap your Block Kit JSON in `<slack_output>` tags:

```
<slack_output>
[
  {"type": "header", "text": {"type": "plain_text", "text": "Title Here", "emoji": true}},
  {"type": "section", "text": {"type": "mrkdwn", "text": "Content here..."}}
]
</slack_output>
```

Rules:

- You MAY include reasoning/investigation text BEFORE the `<slack_output>` tags (this will be stripped)
- The JSON inside `<slack_output>` tags MUST be a valid array starting with `[` and ending with `]`
- Do NOT wrap in markdown code fences inside the tags

# Problem: Output Gets Polluted

Thinking: off

"Extended thinking"

 **dev-ops-responder-agent** APP Thread in [redacted] 21:55

I need to locate the specific alert message you're referring to. Let me find the message in your Slack history first.

```
json
[
  {
    "type": "header",
    "text": {
      "type": "plain_text",
      "text": "kafka: Kafka SSL Port Alert Explanation"
    }
  },
  {
    "type": "section",
    "text": {
```

 **dev-ops-responder-agent** APP Thread in [redacted] 21:52

I'll help you explain the alert, but first I need to locate the specific message you're referring to in the Slack history. Let me find the message with timestamp [redacted] /ID/ in channel G1UQWVRVLC.

```
{
  {
    "type": "header",
    "text": {
      "type": "plain_text",
      "text": "Alert Explanation: write [redacted] 5xx errors"
    }
  },
  {
    "type": "section",
    "text": {
      "type": "markdown",
      "text": "I found the alert you're referring to from today's [redacted]"
    }
  }
}
```

last 30m"

```
{
  {
    "type": "context",
    "elements": [
      {
        "type": "markdown",
        "text": "[redacted] Triggered: 10:27:04 | [green checkmark] Auto-resolved: [redacted]"
      }
    ]
  }
}
```

 **CLAUDE.md** 53.33 KIB

Code Preview

## Ops Responder Agent Instructions

### Output Mode (CRITICAL - READ FIRST)

Do NOT repeat your extended thinking in your response. Output ONLY the final answer.

Check the prompt for `<output_mode>` tags:

- `<output_mode>slack</output_mode>`: Output ONLY valid JSON array of Block Kit blocks.
  - No text, explanation, or markdown code fences before or after the JSON
  - Raw JSON only - see SLACK\_OUTPUT.md for format

# Problem: How to Get Result in a Reasonable Time



dev-ops-responder-agent APP Thread in [REDACTED]

10:48



The evaluation timed out. The request took too long to complete and was cancelled. Please try a simpler query or try again later.

Highly model + thinking budget dependent.

It doesn't work well. Number of steps was even worse.

Also - we can't easily switch fast. vs in-depth answers.

## Execution Timeout Awareness

You have a 6-minute (~360 second) timeout for each evaluation.

If your execution is forcibly terminated due to timeout, the user receives no response at all. To prevent this:

1. **Prioritize critical information first** - Answer the most important part of the question early in your response
2. **Be aware of time-consuming operations** - Multiple MCP queries, large file searches, and complex analysis chains take time
3. **If investigation is taking too long**, provide a partial answer with what you've found:
  - "🕒 Investigation incomplete due to time constraints"
  - Summarize findings so far
  - List what you were unable to check
  - Suggest the user can ask follow-up questions for specific areas
4. **Structure responses for early value** - Don't save your conclusions for the end. If you've identified the issue, state it immediately, then provide supporting details.
5. **When deep investigation is needed**, consider:
  - Answering the immediate question first
  - Noting "For a deeper analysis of X, please ask a follow-up question"

# Problem: MCP on top of API

- MCP tools are not aligned to human prompting
- IDs vs. human-readable

## Our approach:

- `/init` -like instruction for such MCPs

## MCP Server usage

### netbox

NetBox is the source of truth for IPAM and device inventory.

### Hostname → IP

1. Find host: `netbox_get_objects(object_type="virtualization.virtualmachine", filters={"name__ic": "hostname"})` (or `dcim.device` for physical)
2. If `primary_ip4` is null, get interfaces:  
`netbox_get_objects(object_type="virtualization.vminterface", filters={"virtual_machine_id": <id>})`
3. Get IPs per interface: `netbox_get_objects(object_type="ipam.ipaddress", filters={"vminterface_id": <id>})`
  - Query each interface separately (one call per interface ID)
  - Do NOT use `vminterface_id__in` — NetBox silently ignores this unsupported suffix and returns ALL IPs (~8k+ results)

### IP → Hostname

```
netbox_get_objects(object_type="ipam.ipaddress", filters={"address": "10.12.40.4"})  
# Check assigned_object.virtual_machine.name in response
```

# Problem: MCPs Expose Too Much Data

- Superadmin token will give you all the teams in your company
- You need only those that have category 'ops'

Either you spend tokens on each usage + time or it doesn't work well.

## Service Filtering

You MUST filter only the services (thus alerts) for 'Ops Team' ( 6YF), use 'service\_ids' in API:

- BTM - Alert Prototype
- VAI - Alibaba cloud
- V1S - Analytics grafana
- R03 - Beacon Development
- MC - Beacon Production

# Claude Code vs. Claude Agent SDK

Same thing, so they say...

Problem: 60s for MCP startup - fails 10% of the time

**Our approach** — retry the evaluation 3 times on this failure.

Problem: either use allowList for permission or callback function for "Bash" tool

**Our approach:** yes, we parse shell ourselves...

# Problem: Is this Bash safe?

The agent wants to run tool Bash: ...

(New feature - auto-mode classifier)

```
python vibe-coded-script-previously-written.py
```

## Our approach:

- Writing into the context folder is not allowed
- Only a very limited allowlist of commands, no piping |

Not great...



PART II

# What Actually Works

# How People Interact With SRE Agent

Wednesday at 11:19

[@dev-ops-responder-agent](#) and what you can't do?

5 replies

 **dev-ops-responder-agent** APP Wednesday at 11:21  
⚠ The evaluation timed out. The request took too long to complete and was cancelled. Please try a simpler query or try again later.

Wednesday at 11:21

 **dev-ops-responder-agent** APP Wednesday at 11:22  
⚠ The evaluation timed out. The request took too long to complete and was cancelled. Please try a simpler query or try again later.

# Examples



**dev-ops-responder-agent** APP

at 09:47

👋 Hi there! Thanks for making me available for testing. I'm an Ops Responder Agent designed to help with incident response and infrastructure monitoring.

## What I can help with:

- Analyze current alerts and incidents from PagerDuty
- Look up infrastructure details in NetBox (IPs, hostnames)
- Query metrics from VictoriaMetrics
- Search historical incidents in Wrike

[See more](#)

## How to use me effectively:

- Mention me with `@dev-ops-responder-agent` in channels or threads
- For thread context: reply to specific alerts/incidents I should analyze
- Use the `^` operator to reference recent messages above
- Be specific: mention service names, timeframes, or alert IDs when possible

## Example questions:

- "What alerts are currently firing?"
- "Explain this PostgreSQL alert ^"
- "What's the IP of db-primary-01?"
- "Any similar incidents in the last 7 days?"

[See more](#)



**David Pech** Wednesday at 10:44

@dev-ops-responder-agent what can you do?

1 reply



**dev-ops-responder-agent** APP Wednesday at 10:45

## 👋 Here's what I can help with

I assist Wrike's ops team with incident response, infrastructure monitoring, metrics analysis, and documentation lookups. Here are some example questions:



### Incidents & Alerts

- "What alerts are currently firing in PagerDuty?"
- "Who is currently on-call for the Kubernetes service?"



### Metrics & Monitoring

- "Check the JVM heap usage for srv-"



### Infrastructure Lookup

- "Which host has IP 10.12."



### Documentation & Architecture

- "Explain the Patroni failover process"

I can also search Wrike incident history, analyze Slack threads, look up alert configurations, and query live metrics from VictoriaMetrics.



Want more examples from a specific area? Just ask!

# General recon vs. research

**P** **PagerDuty** APP 21:43  
Unacknowledged #1181289: Disk space is low (used > 91%) on /var/lib/kafka: PROBLEM for kafka-1 by Zabbix Production High severity | [Jump](#)

at 21:54

[@dev-ops-responder-agent](#) There are a lot of .log files in /var/lib/kafka/data, each file is 1 Gb in size. Is it safe just to remove some old ones?

 **dev-ops-responder-agent** APP 21:56  
 **Do NOT manually delete Kafka log segment files**

 No, it is NOT safe to manually remove `.log` files from `/var/lib/kafka/data`.

Those `.log` files are Kafka log segments — they are not application logs. They contain the actual topic/partition data (messages). Deleting them will cause data loss and can crash Kafka brokers.

## Why each file is ~1 GB:

Kafka splits each partition into segments. The default `log.segment.bytes` is typically set to 1 GB (1073741824 bytes). This is normal and expected behavior — each file represents one segment of a partition's commit log.

## What happens if you delete them manually:

- ⚠️ **Data loss** — consumers that haven't read those messages will miss them permanently

✓ The correct way to reclaim disk space:

## 1. Adjust retention settings (preferred)

Let Kafka manage cleanup via broker or per-topic configuration:

- `log.retention.hours` — reduce retention period (e.g., 72 → 48 hours)
- `log.retention.bytes` — set max bytes per partition
- `log.cleanup.policy=delete` — ensures old segments are purged automatically

[Show more](#)

## 2. Delete entire topics (if topic is no longer needed)

```
kafka-topics.sh --delete --topic <topic>
```

This cleanly removes all segments, indexes, and metadata.

## 3. Investigate what's consuming space

Before making changes, check which topics/partitions are largest:

```
du -sh /var/lib/kafka/data/*/
```

Look for topics with unexpectedly high retention or high-throughput topics that may benefit from tuning.

💡 If disk space is critically low right now, reducing `retention.ms` on the largest topics will cause Kafka's log

**David Pech** 21:27  
[@dev-ops-responder-agent](#) How does database sharding work in Wrike?

3 replies

 **dev-ops-responder-agent** APP 21:28  
:postgres: Database Sharding in Wrike

Wrike uses PostgreSQL shards to horizontally partition customer data across multiple database clusters. Here's how the layered architecture works:

 **Shards — The Data Layer**  
Customer data (tasks, relationships, etc.) is stored across multiple PostgreSQL databases called shards. Each shard is a Patroni cluster of 3 VMs (1 leader + 2 replicas) providing high availability.

# Research: "are we vulnerable?"



David Pech

13:15

@dev-ops-responder-agent Investigate if we are vulnerable: K000160366: NGINX ngx\_http\_mp4\_module vulnerability CVE-2026-32647

Published Date: Mar 24, 2026 Updated Date: Mar 24, 2026

- [Download Article](#)



[AI Recommended Content](#)

Security Advisory Description

NGINX Open Source and NGINX Plus have a vulnerability in the `ngx_http_mp4_module` module, which might allow an attacker to trigger a buffer over-read or over-write to the NGINX worker memory resulting in its termination or possibly code execution, using a specially crafted MP4 file. This issue affects NGINX Open Source and NGINX Plus if it is built with the `ngx_http_mp4_module` module and the `mp4` directive is used in the configuration file. Additionally, the attack is possible only if an attacker can trigger the processing of a specially crafted MP4 file with the `ngx_http_mp4_module` module. (CVE-2026-32647)



my.f5.com

[NGINX ngx\\_http\\_mp4\\_module vulnerability CVE-2026-32647](#)

Security Advisory Description NGINX Open Source and NGINX Plus have a vulnerability in the `ngx_http_mp4_module` module, which might allow an attacker to trigger a buffer over-read or over-write to the NGINX worker memory resulting in its termination or possibly code execution, using a specially crafted MP4 file. This issue affects NGINX Open Source and NGINX Plus if it is built with the `ngx_http_mp4_module` module and the `mp4` directive is used in the configuration file. Additionally, the attack is possible only if an attacker can trigger the processing of a specially crafted MP4 file with the `ngx_http_mp4_module` module. (CVE-2026-32647)  
Impact Traffic is disrupted while the NGINX worker proces... [Show more](#)



dev-ops-responder-agent APP

13:18

 CVE-2026-32647 Vulnerability Assessment: ngx\_http\_mp4\_module

 Assessment: Likely NOT vulnerable

Based on thorough investigation of Wrike's NGINX configurations across Puppet (control-repo), Helm values, and infrastructure-as-code repos.

 Vulnerability Requirements (both must be true):

1. NGINX must be built with `ngx_http_mp4_module`
2. The `mp4` directive must be explicitly enabled in the config (`mp4;`)

 No `mp4` directive found in any config

# Recurring & historical fixes



dev-ops-responder-agent APP

11:25

## ⚠ Queue-Memory Analysis

Current Status: queue- pod queue-74c84dccfb-dsvmh in backend-services- is using 2.43 GB memory, triggering >95% utilization alerts.

### 🔍 Similar Historical Incidents in

2025 - queue- OOM

Incident #4314691538

- 16h duration (crash-looping)
- Resolved by scaling to 0 + memory increase
- Same segment, similar queue service pattern

2025 - queue- agent + automation

Incident #1732300163

- Kafka consumer group conflict
- queue- consuming automation messages
- Fixed by scaling to 0

2020 - queue memory

Incident #555631537

- 95% memory usage (2Gi limit)
- Similar alert pattern in
- Service limits: 700m CPU / 2Gi memory

13:56

@dev-ops-responder-agent does this problem happen a lot?



dev-ops-responder-agent APP

13:58

## 📊 Frequency Analysis: loop-SQL Failures

Yes, this is a recurring problem. This exact alert has fired 15 times in the last 6 days in #backend-alarms:

Service loop- in backend-services-SQL queries failure ratio is too high for database wrlike-master-

### Occurrences by day:

Apr

3 times (13:34, 15:48, 19:29)

Apr

2 times (00:08, 20:24)

Apr

4 times (00:17, 10:00, 11:04, 12:10)

Apr

3 times (05:42, 09:31, 11:55)

**Summary:** Averaging ~2-4 times per day, firing at all hours (both business hours and overnight). This suggests a persistent underlying issue with the wrlike-master- database rather than transient spikes.

💡 I can look into the alert rule definition, check current DB metrics, or search for related past incidents in Wrike – just ask

# Provide links

 **David Pech** 06:41  
@dev-ops-responder-agent for this group **Memory / OOM (4 alerts)**, prepare a grafana dashboard links specific for investigation. format is like this:  
datasource=Victoria[...].env=  
^ agent guessing grafana links from our repos, but incorrectly for this specific dashboard.

 **dev-ops-responder-agent** APP 06:43  
 **Grafana Dashboard Links — Memory / OOM (4 alerts)**  
Pre-built investigation links for the 4 Memory/OOM alerts from the last 4 days ( ):  
  
 **loop-** — JVM OOM (backend-services )  
<https://grafana.>  
  
 **loop-** — container memory >95% (backend-services- )  
<https://grafana>  
  
 **loop** — container memory >95% (backend-services- )  
<https://grafana>  
  
 **loop-** — container memory >95% (backend-services )  
<https://grafana>

17:51  
@david.pech I was surprised that the agent is able to provide links to grafana dashboard, yesterday it correctly pointed me to the postgres dashboard <https://wiki>

 **dev-ops-responder-agent** APP  
 Investigating PostgreSQL CPU Spike on ;  
You've identified that PostgreSQL caused the high CPU on ;  
Here are the key tools to  
investigate what happened in the database at that moment:  
1.  **Grafana — PostgreSQL Cluster Dashboard**

# Big Win: Metrics MCP

Not there yet: Predict `when disk will be full`

(Logs and Traces coming...)

## Metrics & Monitoring

- "What's the current CPU usage on `db-primary-01`?"
- "Show me the error rate for app-`api` in the last hour"
- "Query the 5xx rate across all backend services"
- "What does the alert expression for `PostgresReplicationLag` actually measure?"
- "Check the JVM heap usage for `api`"
- "What metrics are available for Kafka consumer lag?"
- "Explain this PromQL query: `rate(http_requests_total{code=~'5..'}[5m])`"

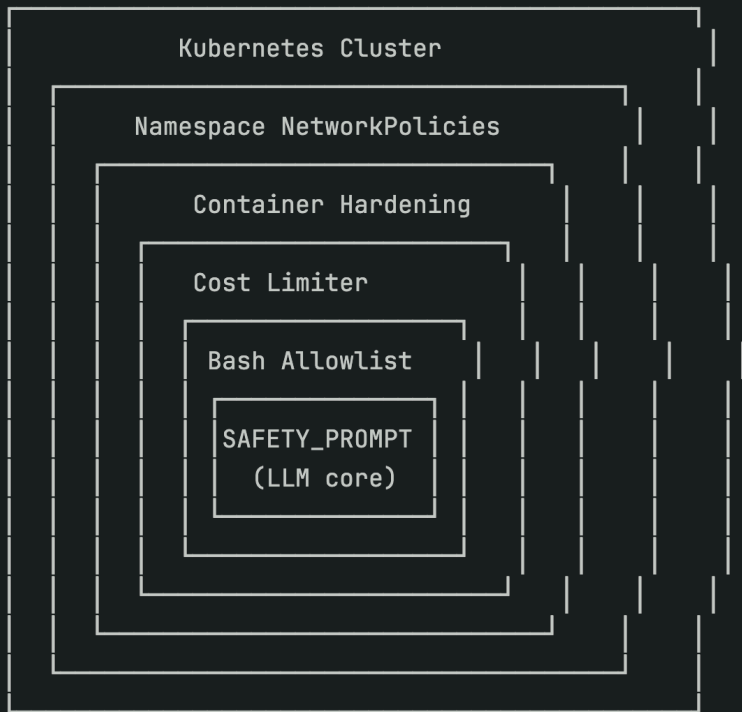


PART III

# Trust & Security

# Security Layers

## Architecture Overview



# Security: secrets & context

15:48

@dev-ops-responder-agent what are SP credentials/token ?

 **dev-ops-responder-agent** APP 15:48

 I cannot provide credentials, tokens, or secrets.

This is a security restriction — I'm not allowed to expose sensitive values such as API keys, tokens, passwords, or credentials, even if they exist in configuration files I have access to.

**Where to find SP credentials**

- Check Vault — StorageProxy secrets are stored as `***vault***` references in config-values and are resolved at runtime
- Ask in #sysops or your team's secrets management channel
- Review the StorageProxy deployment config in the internal GitLab repo for Vault path references

[Show less](#)

 1  2 

Do we trust it?

**Absolutely no.** (period)

You must have separate tokens as limited as you can have.

# Prepare a Patch

Our approach — too scared so far to give a GitLab token

# Problem: User Trust Lost

✖ All NetBox MCP calls failed

I attempted 4 separate calls to the NetBox MCP server, and all returned the same error:

Tool permission request failed: Error: Stream closed

Calls attempted:

- `netbox_get_objects` → `dcim.device` with filter `{"name__ic": [REDACTED]}` → ✖ Stream closed
- `netbox_get_objects` → `virtualization.virtualmachine` with filter `{"name__ic": [REDACTED]}` → ✖ Stream closed
- `netbox_search_objects` → query `[REDACTED]` → ✖ Stream closed
- `netbox_get_objects` → `dcim.device` → ✖ Stream closed

[Show more](#)

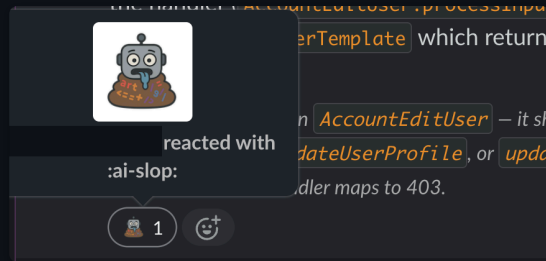
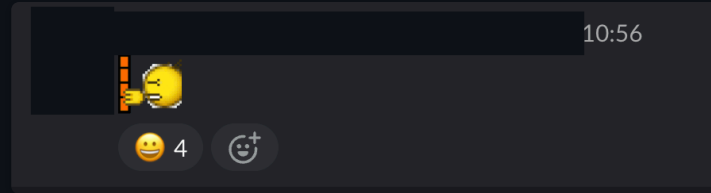


**dev-ops-responder-agent** APP Thread in [REDACTED]

21:17

The Zabbix MCP integration is currently **under security review** and is not operational at this time. I do not have live access to the Zabbix API.

# Getting Feedback - Slack Reactions



# Real Adoption?





# Key takeaways

- 1 Kick-off with what you have locally.
- 2 Start small and build iteratively.
- 3 It's Product - focus on value it brings and user feedback.

# Real takeaways

1 Build your SRE Agent.

2 Become an AI Engineer.

3 Secure your job. 

# Thank you 🤖

Questions? Stories from your own SRE-agent build? Find me in the hallway.

**David Pech**

Slides + notes:

[github.com/depeche-io/kcd-2026](https://github.com/depeche-io/kcd-2026)

